

Reproduction of Lightning-AI/litgpt: Pythia-410M LoRA fine-tuning and ARC-Easy evaluation

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kv:2610.00016v1 · Published 2026-10-07

Independent reproduction · Other · Natural language processing

Abstract

Ran the README fine-tuning example at the pinned commit: LoRA fine-tuning of EleutherAI Pythia-410M on the Alpaca2k dataset for 2 epochs with litgpt's default LoRA settings, followed by a zero-shot ARC-Easy evaluation of the merged model with litgpt evaluate (lm-evaluation-harness). One NVIDIA RTX 5060 Ti.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Pythia-410M LoRA + ARC-E...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: Pythia-410M LoRA + ARC-Easy

Run	Metric	Final value	Step
1	train_loss	1.437	3800
1	val_loss	1.444	3200
1	arc_easy_acc	0.510522	
1	arc_easy_acc_norm	0.455808	

Run	Inline claim	Value	Phase
1	final_val_loss	1.43	
1	final_val_ppl	4.18	
1	arc_easy_acc	0.510522	
1	arc_easy_acc_norm	0.455808	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	516.1 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	d368cd50c552212e0e6fc3bf0bf4538b5cfa6d7a38db5c43faf4f9d360000fbc	4.9 MB
r1.kvbundle.zip	8edae1dc8e3d360285c6f32fed9e4ff8068778eb38f9cb9cf7d49979a36a1356	501 KB

5 Links and licence

Reproduction of: <https://github.com/Lightning-AI/litgpt/tree/18c931c15c2398621d35371a12d2b1785e855bb6>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: lora, fine-tuning, pythia, llm, arc-easy.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Pythia-410M LoRA + ARC-Easy), kv:2610.00016v1/r1. Data hash ee7e477d5d952aada1405b27d8779bf17729398735990b7e7175b181abe00a94.

<https://kveritas.org/records/2610.00016v1/r1.pdf>